



**HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
OFFICE FOR INTERNATIONAL STUDY PROGRAMS**

RETENTION TIME

Department: Processes and Equipment

Class: CC15HC1

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1. ABSTRACT:

1.1. Experimental purpose:

- Survey retention time of a system of continuous stirred-tank reactors(CSTRs).
- Determine real retention time distribution function of and compare with theoretical retention time distribution function.

1.2. Experimental results:

Maximum absorbance (for 1 reactor): $D_0 = 0,03$

Reactor diameter: $d = 14.8$ (cm)

Water level in the CSTR: $H = 12$ (cm)

Flowrate of water: $v = 0,4$ (litter/min)

Table 1: Raw results

t (min)	D		
	1 tank system	2 tank system	3 tank system
1	0.013	0.005	0.027
2	0.026	0.025	0.02
3	0.013	0.018	0.014
4	0.011	0.008	0.013
5	0.014	0.013	0.016
6	0.003	0.004	0.001
7	0.002	0.011	0.002
8	0.019	0.019	0.005
9	0.006	0.002	0.003
10	0.026	0.002	0.005
11	0.009	0.023	0.012
12	0.004	0.001	0.011
13	0.018	0.001	0.003
14	0.011	0.005	0.002
15	0.002	0.006	0.01
16		0.003	0.003
17		0.004	0.005
18		0.001	0.004
19		0.004	0.004
20		0.001	0.009
21		0.009	0.002
22		0.002	0.002
23		0.001	0.005
24		0.02	0.008
25		0.022	0.008
26			0.004
27			0.01
28			0.009
29			0.01
30			0.004
31			0.005
32			0.003
33			0.005
34			0.004
35			0.01

2. EXPERIMENTAL THEORIES :

2.1. Basic concept:

In the equipment system, different fluid elements will follow different paths. Based on defined retention time distribution function, we can evaluate correlatively about the flows in the equipments, drawbacks when designing such as some areas where propeller cannot reach,... and find ways to overcome thanks to this evaluation.

Study of retention time is a necessary method for comparing equipment based on material flow, from which we can improve, set up the optimal model.

Also, based on retention time distribution function, we can operate optimally and thereby set parameters, control methods as well as optimize the process in the equipment.

The retention time of an element in the system is the time that the element stays in the reactor, or in any equipment to be surveyed. Retention time of an equipment is a probability quantity. So, all retention times are around the average retention time, therefore determining the average retention time is particularly significant.

$$\bar{t}_V = \frac{1}{n} \sum_{i=1}^n t_{Vi} \quad (1)$$

Where: t_{Vi} is retention time of an element i .

With the definition of retention time distribution function $F(t_V) = E(t_V)$:

$$t_V = \int_0^{\infty} f(t_V) t_V \cdot dt_V = E(t_V) \quad (2)$$

$$\bar{t}_V = \frac{\int_0^{\infty} c_I^A(t_V) t_V \cdot dt_V}{\int_0^{\infty} c_I^A(t_V) \cdot dt_V} \quad (3)$$

Or:

With point functions we have:

$$\bar{t} = \frac{\sum_{i=1}^K c_i t_i}{\sum_{i=1}^K c_i} \quad (4)$$

K is equal split.

Average retention time according to volume:

$$\tau = \frac{V_R}{V_M} = \bar{t} \quad (5)$$

With V_R : Volume of fluid in the reactor, litter.

V_M : Flowrate of the input, litter/sec.

If the indicator does not reach the ideal correlation, the above equation is not satisfied (If $\tau > \bar{t}$, the indicator may be absorbed into the wall or some additional details of reactor).

2.2. Methods of studying retention time:

The retention time study can be carried out according to the following methods:

- 1) Determine the composition of components at time t (or τ) of leaving the equipment, determine function $F(t)$ or $F(\tau)$.
- 2) Determine the composition of components at time t (or τ) of still remaining in the equipment, function $I(t)$ or $I(\tau)$.
- 3) Determine the composition of components at time t (or τ) of being out of the equipment, function $f(t)$ or $f(\tau)$.

To survey the performance of a real reactor, we often use the stimulus – response method (marking method). The types of input stimuli and output responses are shown in Figure 1.

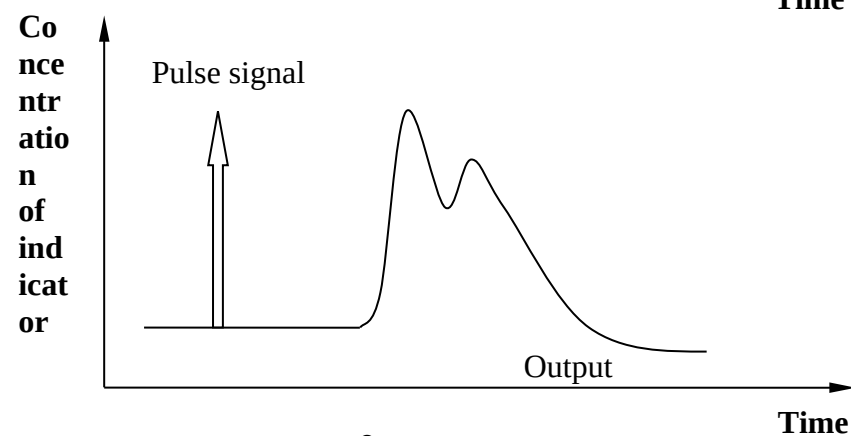
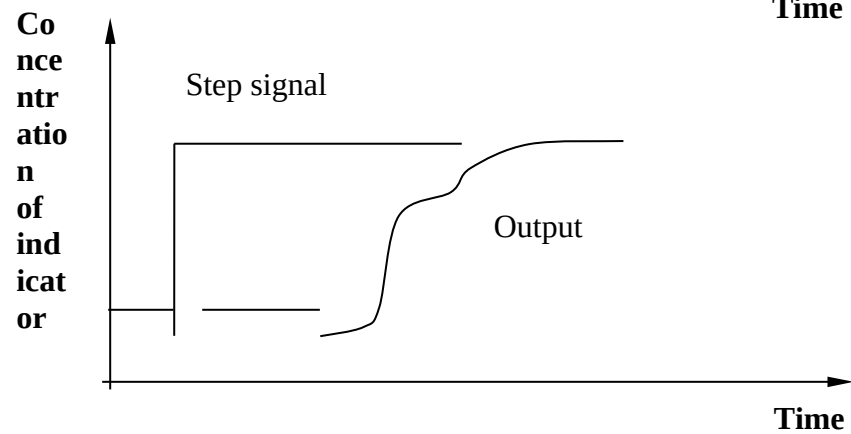
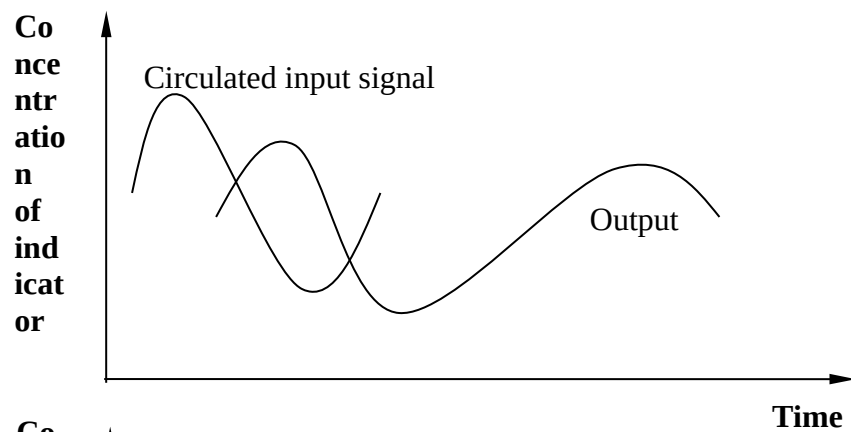
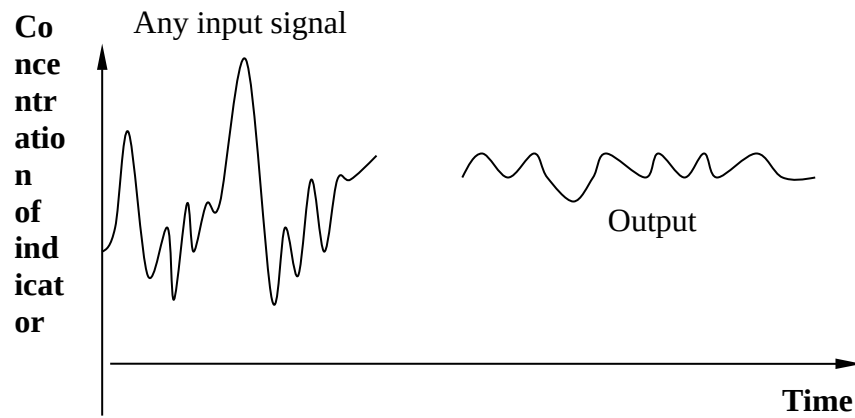
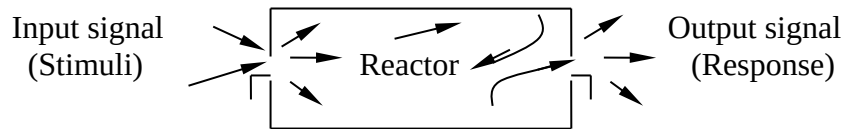


Figure 1: Frequently-used stimulus – response signals

So, marked components must have characteristics that does not influence and distinct from the components that make up the correlation in the system.

Indicators for liquid environment may be colored solution, radioactive substances, stable radioactive isotopes, glowed solid particles... Indicators appropriate to the nature of the components in the system must have appropriate density, viscosity, diffusion coefficient.

When there are appropriate indicators, we can put them into the system in two types of signals: random signal (Stochas) and determined signal (Determinis). Determined signal can be divided into two types: circulated signal and uncirculated signal.

To survey equipment, people use uncirculated determined signal. This type of signal can be created thanks to:

- 1) Marked by collision.
- 2) Marked by continuous input of a specified amount.
- 3) Marked by occupying the whole place in the system.

Because it is convenient for the use and uniformity of the stimulus signal in the form of step and pulse. In this experiment, we use marked by collision type. This type is the implementation of the technical condition of Dirac's function (Dirac's momentum function), also known as Delta's function.

$$\delta(t) = \begin{cases} 0, & t \neq 0 \\ \infty, & t = 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(t) \cdot dt = 1$$

This marked type is suitable for colored solution.

2.3. Retention time distribution function of CSTRs's model:

Most real equipment usually has the distribution function of the CSTRs's model.

One ideal reactor with the type of collision mark must satisfy: Volume V_R in the reactor is constant over time, in the reactor there is a complete agitation of a component in the system homogeneously at all positions in volume V_R . So, in the reactor there is a surge of input flow.

Average retention time according to volume:

$$\tau = \frac{V_R}{V_M} \quad (6)$$

$$\frac{C_I^{ra}}{C_I^{va\theta\theta}} = 1 - \exp\left(-\frac{t}{\tau}\right)$$

When connecting ideal reactors to each other, we have CSTRs's model:

The general form of distribution function:

$$f(t) = 1/\tau = \frac{\left(-\frac{t}{\tau}\right)^{n-1} \cdot \exp\left(-\frac{t}{\tau}\right)}{(n-1)!} \quad (7)$$

When $n = 1$, we have an ideal CSTRs's model. When $n = \infty$, we have an ideal push model.

Assume that initially there was no indicator in the input flow, and then an impulse signal is applied by putting a certain amount of indicator to the fluid flow for a short period of time.

The curve showing the concentration of indicator over time in the output flow corresponding to the pulse stimulation signal at the input is called the curve C. Initial concentration of indicator is C_0 . With the area below the curve is equal to 1, we have:

$$\theta = \frac{t}{t} = \frac{t}{\tau} = \frac{vt}{V} \quad (8)$$

Dimensionless compact time:

With t is a time when any fluid component passes through the equipment.

$\bar{t}$ and τ are determined according to (4) and (5).

$$\int_0^{\infty} C \cdot d\theta = \int_0^{\infty} \frac{C}{C_0} d\theta = 1 \quad \text{or} \quad C_0 = \int_0^{\infty} C \cdot d\theta = \frac{1}{\bar{t}} \int_0^{\infty} C dt \quad (9)$$

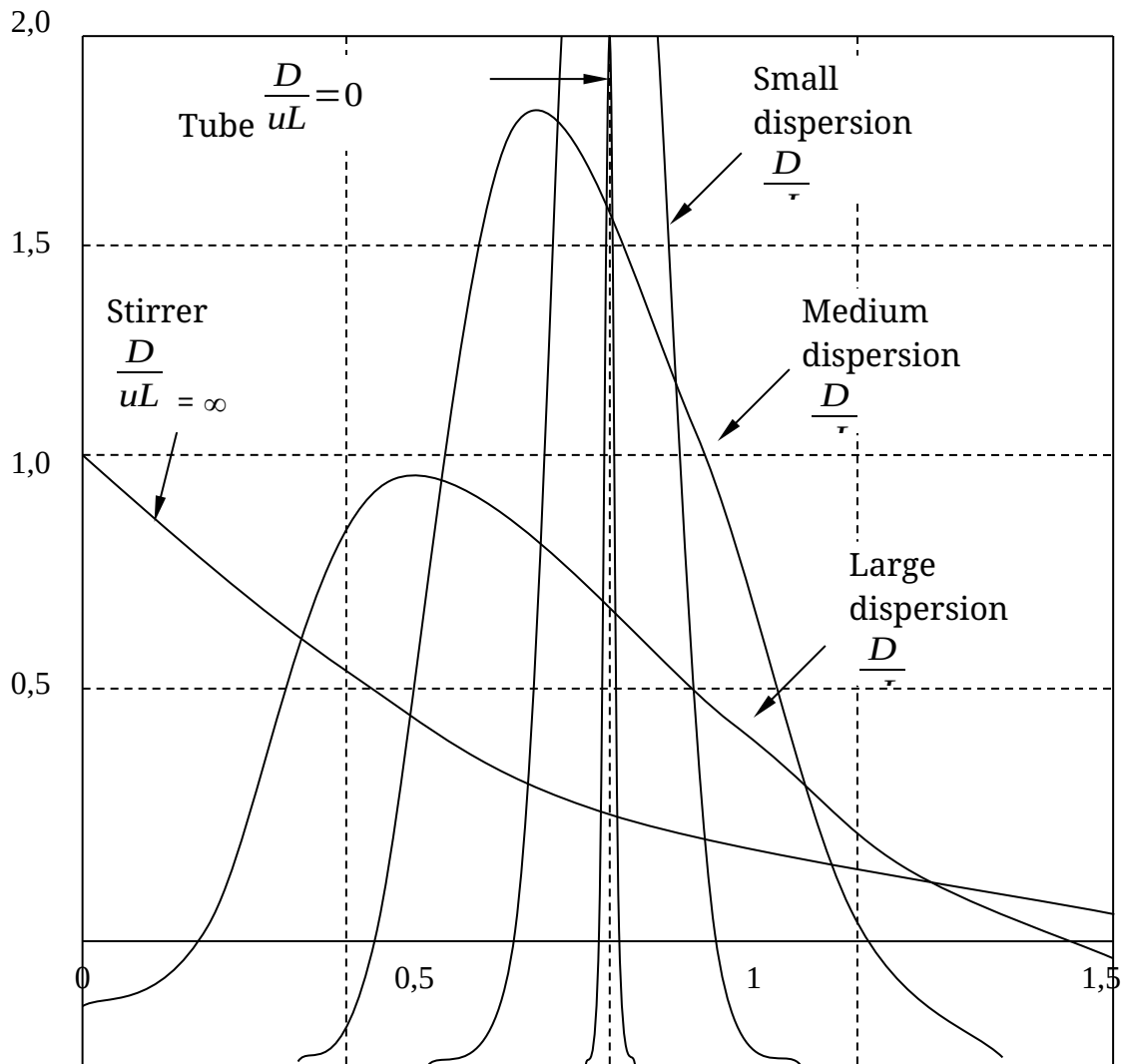


Figure 2: The curve C showing the response at output flow for the pulse signal at the input flow.

3. EXPERIMENTAL EQUIPMENT:

Test tube, stopwatch, pipet, colored substances and light transmittance (absorbance) meter.

4. EXPERIMENTAL METHODS:

Proceed in the following order:

- 1) Open the valve for water to flow into the high-positioned tank until there is water in the overflowing tube.
- 2) Open the valve for water to flow through flow meter into the CSTRs system and adjust the flowrate.
- 3) System of one reactor: When the system is stable, put colored substance into reactor 1. Click the stopwatch (At the same time with the time when we put the colored substance into the reactor), take root time. Use test tubes to take samples from time to time, then compare the color.

- 4) System of two, three reactors: Do the same like one reactor, put the colored substance into reactor 1 and take samples at the last reactor (from the last interconnected pipe).

5. EXPERIMENT RESULTS:

Table 2: 1 tank system

t (min)	D	Experiment			Theory	
		t.D	θ	E	θ	E
1	0.013	0.013	0.14	0.43	0.19	0.82
2	0.026	0.052	0.28	0.87	0.39	0.68
3	0.013	0.039	0.41	0.43	0.58	0.56
4	0.011	0.044	0.55	0.37	0.78	0.46
5	0.014	0.070	0.69	0.47	0.97	0.38
6	0.003	0.018	0.83	0.10	1.16	0.31
7	0.002	0.014	0.97	0.07	1.36	0.26
8	0.019	0.152	1.11	0.63	1.55	0.21
9	0.006	0.054	1.24	0.20	1.74	0.17
10	0.026	0.260	1.38	0.87	1.94	0.14
11	0.009	0.099	1.52	0.30	2.13	0.12
12	0.004	0.048	1.66	0.13	2.33	0.10
13	0.018	0.234	1.80	0.60	2.52	0.08
14	0.011	0.154	1.93	0.37	2.71	0.07
15	0.002	0.030	2.07	0.07	2.91	0.05
Total	0.177	1.281				
t (tb)	7.24				τ	5.16

The values are needed to calculated:

$$D_0 = 0.03 \text{ (m)}$$

$$t_{tb} = 7.24 \text{ (min)}$$

$$v = 0.0004 \text{ (m}^3\text{/min)}$$

$$V = 0.0021 \text{ (m}^3\text{)}$$

$$\tau = 5.16 \text{ (min)}$$

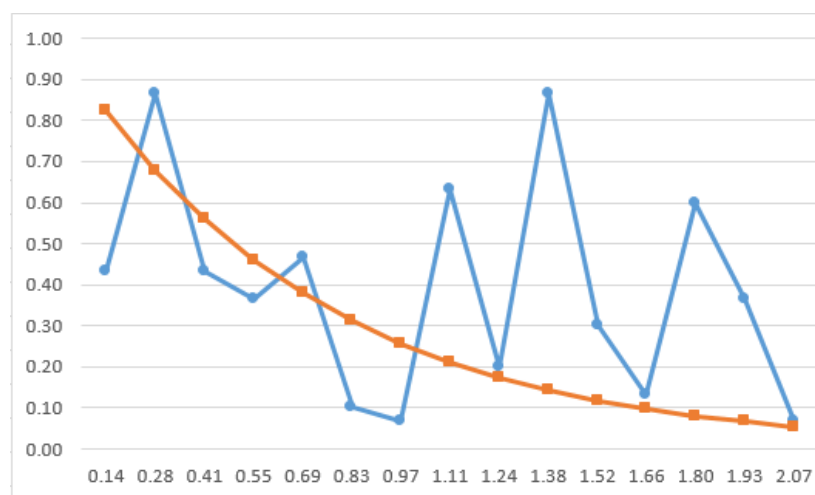


Figure 3: The graph shows the relation $\frac{D}{D_0} = f(\theta)$ for the theoretical and empirical case of a 1-tank system

Table 3: 2 tank system

t (min)	D	Experiment			Theory	
		t.D	θ	E	θ	E
1	0.005	0.005	0.09	0.17	0.10	0.32
2	0.025	0.05	0.17	0.83	0.19	0.53
3	0.018	0.054	0.26	0.60	0.29	0.65
4	0.008	0.032	0.34	0.27	0.39	0.71
5	0.013	0.065	0.43	0.43	0.48	0.74
6	0.004	0.024	0.51	0.13	0.58	0.73
7	0.011	0.077	0.60	0.37	0.68	0.70
8	0.019	0.152	0.69	0.63	0.78	0.66
9	0.002	0.018	0.77	0.07	0.87	0.61
10	0.002	0.02	0.86	0.07	0.97	0.56
11	0.023	0.253	0.94	0.77	1.07	0.51
12	0.001	0.012	1.03	0.03	1.16	0.45
13	0.001	0.013	1.11	0.03	1.26	0.41
14	0.005	0.07	1.20	0.17	1.36	0.36
15	0.006	0.09	1.29	0.20	1.45	0.32
16	0.003	0.048	1.37	0.10	1.55	0.28
17	0.004	0.068	1.46	0.13	1.65	0.24
18	0.001	0.018	1.54	0.03	1.74	0.21
19	0.004	0.076	1.63	0.13	1.84	0.19
20	0.001	0.02	1.71	0.03	1.94	0.16
21	0.009	0.189	1.80	0.30	2.03	0.14
22	0.002	0.044	1.88	0.07	2.13	0.12
23	0.001	0.023	1.97	0.03	2.23	0.10
24	0.02	0.48	2.06	0.67	2.33	0.09
25	0.022	0.55	2.14	0.73	2.42	0.08
Total	0.21	2.451				
t (tb)	11.67				τ	10.32

The values are needed to calculated:

$$D_0 = 0.015 \text{ (m)}$$

$$t_{tb} = 11.67 \text{ (min)}$$

$$v = 0.0004 \text{ (m}^3\text{/min)}$$

$$V = 0.0041 \text{ (m}^3\text{)}$$

$$\tau = 10.32 \text{ (min)}$$

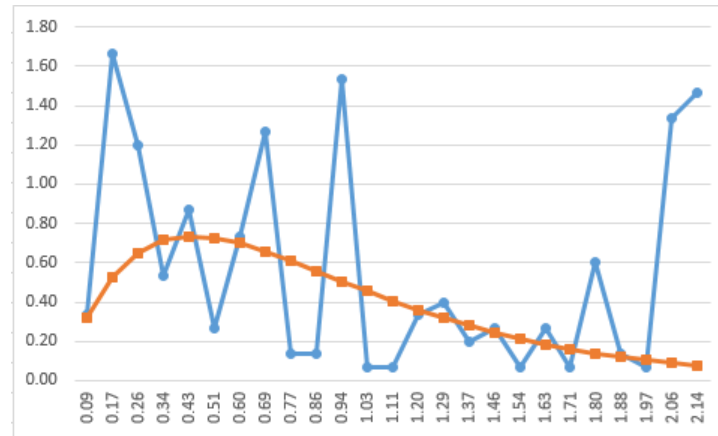


Figure 4: The graph shows the relation $\frac{D}{D_0} = f(\theta)$ for the theoretical and empirical case of a 2-tank system

Table 4: 3 tank system

t (min)	D	Experiment			Theory	
		t.D	θ	E	θ	E
1	0.027	0.027	0.07	0.90	0.06	0.05
2	0.02	0.04	0.13	0.67	0.13	0.15
3	0.014	0.042	0.20	0.47	0.19	0.28
4	0.013	0.052	0.27	0.43	0.26	0.42
5	0.016	0.08	0.33	0.53	0.32	0.53
6	0.001	0.006	0.40	0.03	0.39	0.63
7	0.002	0.014	0.47	0.07	0.45	0.71
8	0.005	0.04	0.54	0.17	0.52	0.76
9	0.003	0.027	0.60	0.10	0.58	0.80
10	0.005	0.05	0.67	0.17	0.65	0.81
11	0.012	0.132	0.74	0.40	0.71	0.81
12	0.011	0.132	0.80	0.37	0.78	0.79
13	0.003	0.039	0.87	0.10	0.84	0.77
14	0.002	0.028	0.94	0.07	0.90	0.73
15	0.01	0.15	1.00	0.33	0.97	0.69
16	0.003	0.048	1.07	0.10	1.03	0.65
17	0.005	0.085	1.14	0.17	1.10	0.60
18	0.004	0.072	1.21	0.13	1.16	0.56
19	0.004	0.076	1.27	0.13	1.23	0.51
20	0.009	0.18	1.34	0.30	1.29	0.47
21	0.002	0.042	1.41	0.07	1.36	0.42
22	0.002	0.044	1.47	0.07	1.42	0.38
23	0.005	0.115	1.54	0.17	1.49	0.35
24	0.008	0.192	1.61	0.27	1.55	0.31
25	0.008	0.2	1.67	0.27	1.61	0.28
26	0.004	0.104	1.74	0.13	1.68	0.25
27	0.01	0.27	1.81	0.33	1.74	0.22
28	0.009	0.252	1.88	0.30	1.81	0.19
29	0.01	0.29	1.94	0.33	1.87	0.17
30	0.004	0.12	2.01	0.13	1.94	0.15
31	0.005	0.155	2.08	0.17	2.00	0.13
32	0.003	0.096	2.14	0.10	2.07	0.12
33	0.005	0.165	2.21	0.17	2.13	0.10
34	0.004	0.136	2.28	0.13	2.20	0.09
35	0.01	0.35	2.34	0.33	2.26	0.08
Total	0.258	3.851				
t (tb)	14.93				τ	15.48

The values are needed to calculated:

$$D_0 = 0.015 \text{ (m)}$$

$$t_{tb} = 11.67 \text{ (min)}$$

$$v = 0.0004 \text{ (m}^3\text{/min)}$$

$$V = 0.0041 \text{ (m}^3\text{)}$$

$$\tau = 10.32 \text{ (min)}$$

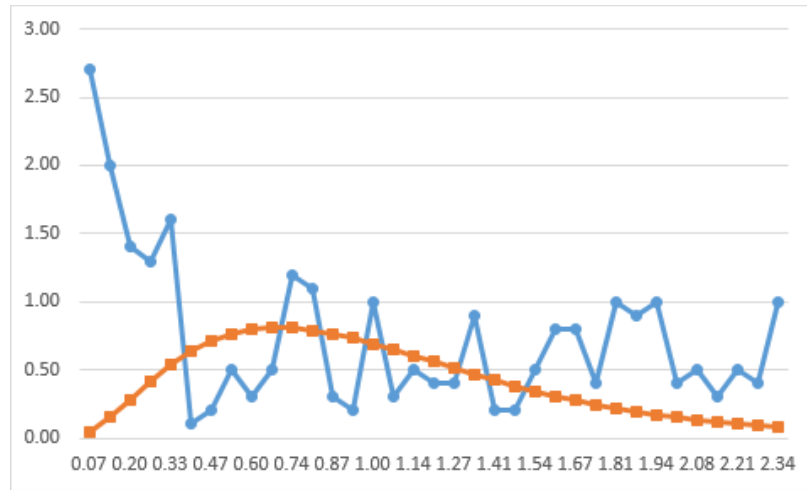


Figure 5: The graph shows the relation $\frac{D}{D_0} = f(\theta)$ for the theoretical and empirical case of a 3-tank system

6. DISCUSSION:

Problem 1: Compare the retention time distribution function of theory and experiment.

◆ 1 tank system: As we could see in the Figure 3, the retention time distribution function of the experiment is totally different compare with the theory. Generally, the E values of the theory is smaller than the experiment. Besides, the compact time values (θ) of theory is larger than the experiment one. That means according to the theory, it might be appear the flow – off or obtuse area. However, the tank does not have lots of fluid pipes so the flow – off phenomena does not happen, just only the obtuse area which the agitators can not affect.

◆ 2 tank system: Following to the Figure 4, the retention time distribution function of the experiment is a little bit look like the theory. That means the E value increases until it's maximum and then decreases. However, the raising is not regular like the theory. It's quite fluctuated. Additionally, the E values of the theory is higher than the experiment.

◆ 3 tank system: The retention time distribution function of the experiment is completely different compare with the theory. It's extremely fluctuated.

Problem 2: The phenomenon of the process and the equipment incures the unsteady

◆ Create the obtuse area in the equipment. The reason is that the speed of the agitators is too slow so there are some places can not be stirred.

◆ Create the circulation of the fluid because of the unevenly mix of the fluid in solution. This may be due to the properties of the fluid or the slow speed of the agitators.

- To overcome this phenomena, we can increase the speed for evenly and all the solution.
- Besides, there are still have some other phenomenon incurring the unsteady:
 - ◆ The alter of the speed and movement direction of the fluid due to the surface shape of the equipment (reactor, agitators, piping, ...). This create some unwanted areas such as swirl area, dead area, flowing area, ..., these places existing in a few time or prolonged can affect the results.
 - ◆ Rotor speed of agitators is unstable over time (due to motor, swirl pivot, ...)
 - ◆ The indicator absorption of equipment makes the concentration of the indicator decreasing over time.
 - ◆ The ambient temperature can alter during experiment. This makes the correlation between the molecules of the fluid changes in the system (viscosity, density, velocity, ...). In addition, it changes the properties of the fluid moving in piping and equipment.

Problem 3: Evaluate the errors

a) Error due to the equipment:

◆ Photometer: This is the biggest reason why the results have the errors

- Because the cuvet and sample are not clean, the dirt is also absorbed the light. Therefore, the results are also be influenced
- The photometer is too sensitive so just a little bit dirty on the cuvet also affect the results. Moreover, depend on the way we put the cuvet into the machine, we get the different values because of the different position of the dirt on the cuvet surface.
- The machine is old may also affect our result.
- The electric supplying for the photometer is not unstable.

◆ Agitators:

- Swirling is unevenly, the velocity is unsteady due to the voltage.
- Swirling slowly makes heterogeneous increasing.

◆ Piping:

Due to the different resistance force of piping and valve, it's difficult to control the flowrate input and output of the tanks. This difficulty increases in case of multi tank system. In reality, we can not keep the volume in the tank compare with beginning during experiment (Eventhough the system works quite stably before adding indicator). During experiment, we have to control the valves so as to the fluid volume in the tank is not alter, so that it will affect the steady of the system.

b) Operation errors:

- Because of the time we determine to get the samples.
- We might not clean the cuvet clearly The cuvet does not dry before measurement (The old sample could be still in the cuvet, the result obviously has the errors). Furthermore, the experiment condition is one of

the important reason (Does not have the apparatus for drying, does not have many replace cuvet).

- The flowrates in the tank are controlled unstably.

c) Errors because of the initial moment and pulse signal:

We can not be established the right equation of the momentum function DIRAC because the time controlling can not be operated for indicator equal zero (In real, the experiment for indicator by pipet lose 5s), this time is quiet long and so that it makes the different between the curve of theory and experiment (is built from momentum function DIRAC).

7. CALCULATION:

Calculated values for 1 tank system

7.1.1. Experiment:

Maximum absorption: $D_o = 0.03$

Average retention time: $\bar{t} = \frac{\sum D_i t_i}{\sum D_i}$

Compact time: $\theta = \frac{t}{\bar{t}}$

Measurement of retention time's distribution: $E_\theta = \frac{D}{D_o}$

7.1.2. Theory:

Water flowrate input to the tank: $v = 0,4 \text{ (l/min)} = 0,4 \cdot 10^{-3} \text{ (m}^3\text{/min)}$

Tank volume: $V = (\pi/4) \cdot 0,148^2 \cdot 0,12 = 0,0021 \text{ (m}^3\text{)}$

Volume time: $\tau = \frac{V}{v} = 5.16$

Compact time: $\theta = \frac{t}{\tau}$

Measurement of retention time's distribution: $E_\theta = e^{-\theta}$

Calculated values for 2 tank system

7.1.3. Experiment:

Maximum absorption: $D_o = 0.03/2 = 0.015$

Average retention time: $\bar{t} = \frac{\sum D_i t_i}{\sum D_i}$

Compact time: $\theta = \frac{t}{\bar{t}}$

Measurement of retention time's distribution: $E_\theta = \frac{D}{D_o}$

7.1.4. Theory:

Water flowrate input to the tank: $v = 0.4 \text{ (l/min)} = 0,4 \cdot 10^{-3} \text{ (m}^3\text{/min)}$

Tank volume: $V = 2 \cdot (\pi/4) \cdot 0,148^2 \cdot 0,12 = 0,0041 \text{ (m}^3\text{)}$

$$\text{Volume time: } \tau = \frac{V}{v} = 10.32$$

$$\text{Compact time: } \theta = \frac{t}{\tau}$$

$$\text{Measurement of retention time's distribution: } E_{\theta} = \frac{N(N\theta)^{N-1}}{(N-1)!} e^{-N\theta} = 2 \cdot (2\theta) e^{-2\theta}$$

Calculated values for 3 tank system

7.1.5. Experiment:

$$\text{Maximum absorption: } D_o = 0.03/3 = 0.01$$

$$\text{Average retention time: } \bar{t} = \frac{\sum D_i t_i}{\sum D_i}$$

$$\text{Compact time: } \theta = \frac{t}{\bar{t}}$$

$$\text{Measurement of retention time's distribution: } E_{\theta} = \frac{D}{D_o}$$

7.1.6. Theory:

$$\text{Water flowrate input to the tank: } v = 0.4 \text{ (l/min)} = 0.4 \cdot 10^{-3} \text{ (m}^3\text{/min)}$$

$$\text{Tank volume: } V = 3 \cdot (\pi/4) \cdot 0.148^2 \cdot 0.12 = 0.0062 \text{ (m}^3\text{)}$$

$$\text{Volume time: } \tau = \frac{V}{v} = 15.48$$

$$\text{Compact time: } \theta = \frac{t}{\tau}$$

$$\text{Measurement of retention time's distribution: } E_{\theta} = \frac{N(N\theta)^{N-1}}{(N-1)!} e^{-N\theta} = \frac{3(3\theta)^2}{2} e^{-3\theta}$$

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